Exercise 2: Machine Learning Paradigms & Data Visualization

Task 1: Another quiz!

Identify the general type of machine learning issues listed below (types: classification, clustering, regression, generation):

- House prices: Given past sales data, determine the expected price of a newly listed property.
- Buying habits: Analyze purchase histories to discover groups of people with comparable buying habits.
- Banking transactions: Based on past banking data, determine whether a new transaction is suspicious.
- Human faces: Develop realistic-looking portraits of people who do not actually exist.
- Predicting electricity demand: Use past consumption records to determine future energy usage patterns.
- Movie reviews: Analyze movie reviews and predict the expected star rating (1 to 5).
- Real Estate: An algorithm estimates the market price of a house based on square footage and location.
- Credit Check: A bank wants to determine if a loan applicant is "High Risk" or "Low Risk."
- Animal Recognition: An app analyzes a photo to determine if it’s a Golden Retriever, a Poodle, or a Beagle.
- Weather Forecast: A model predicts the exact amount of rainfall (in millimeters) for the upcoming weekend.
- Medical Screening: A system analyzes an X-ray to detect whether a bone is "Fractured" or "Not Fractured."
- Genetics: A researcher uses a model to group genes with similar expression patterns to find new, unknown biological functions.

Task 2: Choosing the Right Visualization

Please download the Jupyter Notebook from our Moodle course.
Complete the tasks directly within the Jupyter Notebook.
The focus is on selecting the correct plot types to visualize specific data relationships and identifying why certain visualizations are misleading or ineffective.